

Business Intelligence – 5 Minute Revision Sheet

1. Business Intelligence (BI) – Definition

BI is a set of technologies, processes and tools used to collect, analyze and present business data for better decision-making.

Converts raw data → meaningful insights → strategic decisions.

2. Brief History of BI

1958 – Hans Peter Luhn introduced the concept of Business Intelligence.

1970s–80s – Decision Support Systems (DSS).

1980s–90s – Data Warehousing and OLAP.

1990s–2000s – BI tools and reporting systems.

2010s–Present – Cloud BI, AI integration, real-time analytics.

3. BI Architecture (Simple Flow)

Data Sources → ETL → Data Warehouse → Data Marts → OLAP/Data Mining → Reports & Dashboards

Data Sources: POS, CRM, ERP, web data.

ETL: Extract, Transform, Load data for consistency.

Warehouse: Central storage for integrated historical data.

4. BI Scenarios (Applications)

Sales analysis – identify best selling products.

Customer behavior analysis – personalize marketing.

Financial monitoring – track profit and expenses.

Supply chain optimization – manage inventory.

Fraud detection – identify suspicious transactions.

5. Future & Goals of BI

Integration with AI and Machine Learning.

Real-time analytics and predictive insights.

Cloud-based BI platforms.

Self-service BI for non-technical users.

Goals: better decisions, efficiency, trend detection, competitive advantage.

6. Data vs Information vs Knowledge

Data – raw facts (e.g., sales numbers).

Information – processed data (monthly sales report).

Knowledge – insights used for decisions (sales increase during festivals).

Flow: Data → Processing → Information → Analysis → Knowledge → Decision.

7. BI Tasks

Data collection – gather business data.

Data integration – clean and combine data.

Data storage – warehouse or marts.

Data analysis – OLAP, statistics, data mining.

Visualization – dashboards and reports.

8. Analysis Formats

Reporting – structured summaries.

Ad-hoc queries – custom analysis.

OLAP – multidimensional analysis (drill-down, roll-up).

Statistical analysis – trends and correlations.

Data mining – pattern discovery.

9. Decision Support System (DSS)

DSS is a computer-based system that helps managers analyze data and make decisions.

Supports semi-structured and unstructured problems.

Components: Database + Model base + User interface.

10. Information Systems for Decision Making

TPS – records daily transactions.

MIS – periodic reports for managers.

DSS – analytical decision support.

EIS – strategic information for executives.

BI – advanced analytics and dashboards.

11. BI & DSS Connection

BI collects and analyzes business data.

DSS uses BI insights + analytical models for decision making.

BI = insight generation, DSS = decision evaluation.

12. Simon's Decision-Making Model

Proposed by Herbert Simon.

Four phases:

- 1) Intelligence – identify problem.
- 2) Design – develop possible solutions.
- 3) Choice – select best option.
- 4) Implementation – execute decision.